THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $(x+3)^2$ equals: 1

(a) x^2+6x **(b)** x^2+9 **(c)** x^2+6x+9 **(d)** x^2+3x+9

2. $(x-2)^2$ equals: 1

(a) x^2-4x+4 **(b)** x^2-2x+4 **(c)** x^2-4 **(d)** x^2+4x+4

3. $(x+3)(x-3)$ equals: 1

(a) x^2-9 **(b)** x^2+6x-9 **(c)** x^2-6x+9 **(d)** x^2+9

4. $(x+2)(x+5)$ equals: 1

(a) $x^2+7x+10$ **(b)** $x^2+10x+7$ **(c)** x^2+7x+7 **(d)** x^2+10

5. 101×99 equals: 1

(a) 9998 **(b)** 10001 **(c)** 10000 **(d)** 9999

6. In the expansion of $(2x+3)(4x+5)$ the coefficient of x is: 1

(a) 10 **(b)** 12 **(c)** 22 **(d)** 23

7. Which of the following is true for all values of a and b? 1

(a) $(a+b)(a-b)=a^2+b^2$ **(b)** $(a-b)^2=a^2-2ab+b^2$ **(c)** $(a+b)^2=a^2+b^2$ **(d)** $(a-b)^2=a^2-b^2$

8. $(x+4)^2$ equals: 1

(a) x^2+16 **(b)** $x^2+8x+16$ **(c)** $x^2+4x+16$ **(d)** x^2+8x

9. $(x-5)^2$ equals: 1

(a) $x^2-10x+25$ **(b)** $x^2+10x+25$ **(c)** $x^2-5x+25$ **(d)** x^2-25

10. $(x+6)(x-6)$ equals: 1

(a) $x^2-12x+36$ **(b)** $x^2+12x-36$ **(c)** x^2-36 **(d)** x^2+36

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Expand $(x+8)^2$ using an identity. 2

12. Expand $(2x+1)(3x+4)$. 2

13. Find 48×52 using $(a+b)(a-b)$. 2

14. Expand $(x + 9)^2$ using an identity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Expand $(2x + 3)^2$. 3

16. Find 53×47 using the identity $(a + b)(a - b) = a^2 - b^2$. 3

17. Simplify $(x + 2)(x + 3) + (x - 2)(x - 3)$. 3

OR

Expand $(3x + 4)^2$. 3

18. Find 68×72 using the identity $(a + b)(a - b) = a^2 - b^2$. 3

OR

Simplify $(x + 4)(x + 5) + (x - 4)(x - 5)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangle has length $(x + 3)$ cm and breadth $(x + 5)$ cm. Write an expression for its area and find the area when $x = 4$. 5

OR

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ and verify it for $a = 7$, $b = 3$. 5

20. A rectangle has length $(x + 2)$ cm and breadth $(x + 6)$ cm. Write an expression for its area and find the area when $x = 5$. 5

OR

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ and verify it for $a = 9$, $b = 4$. 5

END OF QUESTION PAPER — Total: 40 Marks

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